

EDUCATION

B.Sc. in Electrical Engineering — University of Tehran

Faculty of Electrical & Computer Engineering · Ranked #1 in Iran, QS World University Rankings 2026

Sep 2022 – Sep 2026

GPA **18.17 / 20** · ranked **13th of 128** — top 10% of cohort

Thesis: *Improving Spatial Focality in Temporal Interference Deep Brain Stimulation via Phase Modulation* — supervised by Prof. Hoda Ameri · jury Prof. Amideddin Mataji-Kojouri · grade **20/20**

Filter & Circuit Synthesis	20	Engineering Mathematics	20	Linear Control Systems	20
Electronics II	20	Microwave I	19.8	Signals & Systems	19.74
Microwave Lab	19.54	Antenna I	19.5	Wireless Communications	19.4
Electric Circuits I	19	Principles of Communication Systems	18.8	Electronics I	18.5
Communication Circuits	18.5	Electromagnetics	18.26	Fields & Waves	18.1
Electromagnetic Compatibility (EMC)	17.75				

RESEARCH EXPERIENCE

Bioelectromagnetics Laboratory — University of Tehran

Jun 2025 – Nov 2025

Undergraduate research intern · Profs. R. Faraji-Dana, H. Ameri & A. Mataji-Kojouri

- Developed a non-invasive deep brain stimulation technique that targets deep neural circuits without surgical implants.
- Proposed **PM-TI**, a phase-modulated temporal-interference method that reduces unintended stimulation spread by $\approx 70\%$, sharpening spatial focus.
- Simulated electric-field distributions in biological phantoms with COMSOL Multiphysics to validate the focused hotspot.
- Built a 2-channel high-precision current source (enhanced Howland architecture) to generate the stimulation signals in hardware.
- Co-authored the PM-TI methodology paper, accepted at ICEE 2026.

PUBLICATIONS

1.

H. Keshtkar

, M. Moltamesi, H. Ameri, A. Mataji-Kojouri, R. Faraji-Dana, “Spatial Focusing of Temporal Interference Deep Brain Stimulation via Phase Modulation,” *34th Int. Conference on Electrical Engineering (ICEE)*, 2026.
ACCEPTED – PENDING IEEE INDEXING

2.

M. Moltamesi

,

H. Keshtkar

, et al., “Seizure Suppression in a Neural Mass Model via Input–Output Feedback Linearization and Backstepping Control,” *34th Int. Conference on Electrical Engineering (ICEE)*, 2026.
ACCEPTED – PENDING IEEE INDEXING

HONORS & AWARDS

- Top **0.5% nationwide** in Iran’s university entrance exam (Konkur, Math & Physics) among 146,000+ participants — 2022.
- Ranked 13th of 128 in the electrical engineering cohort.
- Exam-exempt admission to the M.Sc. program at the University of Tehran, reserved for top-ranked undergraduates.

TEACHING ASSISTANT

Fields & Waves	Prof. Reza Faraji-Dana	University of Tehran	Fall 2026
Electromagnetics	Prof. Hoda Ameri	University of Tehran	Fall 2025
Electromagnetics	Prof. Asghar Keshtkar	Imam Khomeini Int. University	Fall 2025
Electric Circuits I	Prof. Moein Abedini	University of Tehran	Fall 2024

SKILLS

 MATLAB

 Python

 C

 Verilog HDL

 COMSOL Multiphysics

 Ansys HFSS

 Ansys Maxwell

 ADS

 NI Multisim

 Arduino

LANGUAGES

English — professional working proficiency · TOEFL iBT **100/120** (L 6.0 · S 5.0 · R 4.5 · W 3.5) · CEFR C1 · tested May 2026

Persian — native